One Predictor, Two Regimes: Zero-Shot Confusion Matrix Diagonal Strength Predicts Transfer Learning Outcome Before Fine-Tuning

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Abstract

We present evidence that the diagonal strength of a zero-shot confusion matrix, computed from a single zero-shot evaluation on a labeled target set before any fine-tuning, predicts transfer learning outcome in two distinct regimes. On structure-preserving geometric shifts such as rotations and blur, higher diagonal strength predicts a more favorable transfer outcome. On structure-destroying degradations such as Gaussian noise, salt-and-pepper noise, and contrast reduction, lower diagonal strength predicts a larger transfer benefit. The same scalar carries opposite predictive information in the two shift regimes.

In the rotation experiments, the binary prediction correlation was r = 0.445 (p = 0.000006, n = 96) across eight MLP architecture sizes and replicated on blur transforms across two datasets (r = 0.495 and r = 0.564). In the degradation experiments, the magnitude prediction correlation ranged from r = -0.573 to r = -0.941 across four degradation experiments spanning three degradation types and two datasets. We document boundary conditions where the method did not produce useful predictions and report all negative results alongside positive findings. We also report an exploratory secondary signal in off-diagonal entropy and one financial temporal regime-shift validation on Lending Club data, with evaluation on 170,522 held-out test loans. All experiments use real published datasets with no synthetic data.

1 Introduction

Transfer learning is widely used but poorly screened in practice. A practitioner who wants to know whether a source model will help on a new target task must typically fine-tune the model on that task and compare the result against training from scratch. This is expensive, and in many settings most candidate transfers hurt rather than help. In the rotation experiments reported in this paper, only 8 out of 96 tested transfers produced a positive accuracy advantage, meaning 91.7% of fine-tuning runs wasted compute on models that degraded target performance.

This paper asks whether the outcome of transfer learning can be predicted before fine-tuning begins, using only a zero-shot forward pass on the target domain. Specifically, we ask whether the diagonal strength of the zero-shot confusion matrix, defined as the fraction of total predictions that fall on the diagonal, carries useful information about what fine-tuning will produce.

The answer depends on what kind of domain shift is involved. We find two distinct prediction regimes. On structure-preserving geometric shifts such as rotations and blur, higher diagonal strength predicts a more favorable transfer outcome. On structure-destroying degradations such as Gaussian noise, salt-and-pepper noise, and contrast reduction, lower diagonal strength predicts that transfer will provide a larger benefit. The same scalar, computed from the same zero-shot target evaluation, carries opposite predictive information depending on the nature of the shift.

This two-regime finding is the central scientific result of the paper. It is not a claim that the predictor is perfect. The binary regime produces moderate correlations ($r = 0.445$ to 0.564 in our sample) and the magnitude regime produces stronger ones ($r = -0.573$ to -0.941). Both are statistically significant across all validated positive-result conditions reported in this paper. Both also have documented boundary conditions where the method did not produce useful predictions, and those boundaries are reported alongside the positive results.

We also report a secondary finding. Off-diagonal entropy, which measures how randomly the model distributes its mistakes across incorrect classes, is a statistically significant secondary predictor of transfer outcome in the rotation experiments, but only once sample size is sufficient. At $n = 12$ the entropy signal did not reach significance. At $n = 96$ it did ($r = -0.313$, $p = 0.002$). We treat this as an exploratory finding rather than a primary claim.

Finally, we report one financial temporal regime-shift validation on tabular data. The same binary prediction rule correctly identified negative transfer on Lending Club 2013–2016 data, with the performance difference tested on 170,522 held-out target-period loans ($z = 13.24$, $p < 0.000001$). This is a single confirmed directional prediction, not a correlation study, and we frame it accordingly.

The paper makes the following contributions:

- We show that zero-shot confusion matrix diagonal strength predicts transfer learning outcome before fine-tuning, and that the prediction regime depends on the type of domain shift.
- We show that on structure-preserving geometric shifts, higher diagonal strength predicts more favorable transfer outcomes, with this relationship holding across all eight tested MLP architecture sizes spanning hidden layer sizes from 16 to 384 neurons.
- We show that on structure-destroying degradations, lower diagonal strength predicts larger transfer benefit, with this inverse relationship replicating across four degradation experiments spanning three degradation types and two datasets.
- We document boundary conditions where the method did not produce useful predictions, including Fashion-MNIST salt-and-pepper noise ($p = 0.053$), CIFAR-10 Gaussian noise (insufficient diagonal range), and cross-domain pairs with near-zero feature overlap.
- We report an exploratory secondary signal in off-diagonal entropy and a single financial temporal regime-shift validation, both framed with appropriate scope.

All experiments use real published datasets. No synthetic data was used at any point. Negative results are reported throughout.

2 Related Work

Transfer learning. Pan and Yang (Pan and Yang, 2010) provide a foundational survey of transfer learning settings, distinguishing inductive, transductive, and unsupervised transfer. Yosinski et al. (Yosinski et al., 2014) show empirically that the transferability of learned features degrades with distance between source and target domains, and that lower layers transfer more broadly than higher layers. Our work does not study which features transfer but whether a given source model will produce a positive fine-tuning outcome on a specific target domain, which is an earlier decision in the transfer learning pipeline.

Transferability estimation without fine-tuning. The closest prior work to ours is the LEEP score of Nguyen et al. (Nguyen et al., 2020), which estimates transferability from the empirical conditional distribution of target labels given source model predictions, computed in a single forward pass. LEEP requires access to the source model's output probability vector over source classes. Our predictor uses only predicted target labels and ground-truth target labels, rather than probability vectors over source classes. Tran et al. (Tran et al., 2019) study task hardness and transferability across supervised classification tasks using feature geometry. These approaches are complementary to ours: they focus on representation-level signals while our predictor operates on the confusion matrix of zero-shot predictions evaluated on a labeled target validation set, requiring only predicted class labels and ground-truth target labels rather than probability outputs or internal feature access.

Domain shift and covariate shift. Ben-David et al. (Ben-David et al., 2010) establish theoretical bounds on generalization under distribution shift, showing that target error depends on both source error and a divergence term between domains. Our empirical finding that the method requires some structural similarity between source and target domains (as evidenced by the KMNIST to MNIST failure) is consistent with this theoretical framework: when domain divergence is near-total, zero-shot performance carries no useful signal about fine-tuning outcome. Our two-regime taxonomy (geometric shifts versus degradation shifts) can be understood as an empirical characterization of how shift type modulates the direction of this signal.

3 The Diagonal Strength Predictor

3.1 Definition

For a classification model evaluated on a target domain, the zero-shot confusion matrix captures the full pattern of predictions before any target-domain training. The diagonal strength is the fraction of total predictions that land on the correct class:

$$F = \frac{\text{trace}(CM)}{\sum_{i,j} CM_{ij}} \tag{1}$$

where CM is the confusion matrix computed from zero-shot predictions on the target domain. F equals zero-shot accuracy on the target domain and lies in $[0, 1]$.

This is the only quantity needed for the binary prediction regime. For the secondary entropy signal discussed in Section 8, we also compute the off-diagonal entropy from the normalized off-diagonal mass of the same confusion matrix, but that metric is not part of the primary prediction method.

3.2 The Two-Regime Hypothesis

The central hypothesis of this paper is that F changes predictive role depending on the type of domain shift between source and target. We distinguish two shift types.

A geometric shift preserves the basic structure of inputs while changing their spatial arrangement. Rotations and blur fall into this category. A degradation shift reduces input quality by adding noise or reducing contrast. Gaussian noise, salt-and-pepper noise, and contrast reduction fall into this category.

For geometric shifts, a model with high F on the target domain already partially solves the target task at zero shot, suggesting that source features transfer well. Fine-tuning such a model should produce a better outcome than fine-tuning a model with low F.

For the validated degradation shifts in this paper, transfer was positive in nearly all cases because clean pre-training provides better features than degraded data alone. The question is not whether transfer helps but how much. A model with low F is being tested on heavily degraded data, where the gap between pre-trained features and target-domain features is large. That gap is precisely what fine-tuning closes. A model with high F is being tested on lightly degraded data, where fine-tuning provides less incremental benefit because the features are already adequate.

These two regimes predict opposite correlation directions from the same scalar. Sections 5 and 6 test both predictions empirically.

3.3 Transfer Advantage

Throughout this paper, the outcome variable is transfer advantage, defined as:

$$\Delta = \text{accuracy}_{\text{transfer}} - \text{accuracy}_{\text{scratch}} \tag{2}$$

A positive Δ means fine-tuning from a source-trained model outperformed training from scratch on the target domain. A negative Δ means training from scratch was better. All comparisons use a corrected fair-training protocol: the fine-tuned model receives 20 source epochs followed by 20 target epochs, and the scratch model is trained on the target domain for 40 epochs.

4 Datasets and Experimental Protocol

4.1 Datasets

All experiments use real published datasets. No synthetic data was generated at any point.

MNIST (LeCun et al., 1998): 70,000 grayscale handwritten digit images across 10 classes (60,000 train, 10,000 test). Used as both source and target domain across rotation, blur, noise, and contrast experiments.

Fashion-MNIST (Xiao et al., 2017): 70,000 grayscale clothing item images across 10 classes (60,000 train, 10,000 test). Used as source and target domain in blur and Gaussian noise experiments. Texture-based features make this dataset harder than MNIST for the architectures tested.

Lending Club (Lending Club, 2016): Loan origination and outcome records from 2013 to 2016. We use 2013–2014 records as the source period (370,039 loans, 16.9% default rate) and 2015–2016 records as the target period (852,607 loans, 17.7% default rate). This dataset is used only for the temporal regime-shift validation in Section 7.

4.2 Architecture Families

Two architecture sets were used.

MLP architectures: Eight fully-connected networks with hidden layer sizes ranging from Tiny (16, 8) to Massive (384, 192). All MLP experiments use scikit-learn MLPClassifier with ReLU activations.

Five-architecture subsets: For the blur, noise, and contrast experiments, a five-architecture subset was used: Small (32, 16), Medium (64, 32), Large (128, 64), Very Large (192, 96), and Huge (256, 128).

4.3 Experimental Protocol

For each image experiment, the following steps were executed in order:

1. Train a source model on the source domain for 20 epochs.
2. Evaluate the source model zero-shot on the target domain and compute F from the resulting confusion matrix.
3. Fine-tune the source model on the target domain for 20 additional epochs (warm start).
4. Train a scratch model on the target domain from random initialization for 40 epochs.
5. Compute transfer advantage Δ as fine-tuned accuracy minus scratch accuracy on a held-out test set.

Diagonal strength is computed from zero-shot predictions evaluated against labeled target domain examples before any fine-tuning. The total training budget is matched in total training iterations: the fine-tuned model receives 20 source epochs plus 20 target epochs, while the scratch model receives 40 target epochs. Pearson correlation between F and Δ is computed across the full set of architecture-condition pairs within each experiment. P-values are two-tailed and the significance threshold is $p < 0.05$ throughout.

4.4 Audit and Reproducibility

All image experiment results were verified against stored JSON output files. An independent audit confirmed 1,591 mathematical integrity checks across the full image experiment set, with all correlations matching documented values within a tolerance of ± 0.001. The financial experiment passed a separate 31-check data integrity audit. All validation scripts and stored result files are available at `https://github.com/Wise314/identity-framework-extensions`.

5 Binary Prediction on Geometric Shifts

5.1 Setup

We test whether zero-shot diagonal strength predicts transfer advantage on structure-preserving geometric shifts. Two shift types were validated: rotations and blur. In both cases the source and target domains share the same image class structure, with spatial arrangement changed by the transform.

For the rotation experiments, source models were trained on standard MNIST and evaluated zero-shot on MNIST rotated at angles from 15 to 180 degrees in 15-degree increments (12 target conditions). This was run across all eight MLP architecture sizes, producing 96 architecture-condition pairs. For the blur experiments, source models were trained on clean MNIST or clean Fashion-MNIST and evaluated zero-shot on the same dataset with motion blur applied at five kernel sizes (3×3, 5×5, 7×7, 9×9, 11×11), producing 25 pairs per dataset across the five-architecture subset described in Section 4.

5.2 MNIST Rotation Results

The overall Pearson correlation between diagonal strength and transfer advantage across all 96 rotation tests was $r = 0.445$ ($p = 0.000006$). Transfer was positive in only 8 of 96 cases (8.3%). Within this experimental matrix, 91.7% of fine-tuning runs produced negative outcomes, underscoring the practical value of screening transfer candidates before fine-tuning.

Architecture	Hidden Layers	Correlation	P-value	Positive Transfers
Medium	(64, 32)	0.822	0.0010	2/12
Small	(32, 16)	0.822	0.0010	1/12
Large	(128, 64)	0.789	0.0023	1/12
Tiny	(16, 8)	0.758	0.0043	0/12
Medium-Large	(96, 48)	0.685	0.0140	1/12
Huge	(256, 128)	0.676	0.0158	2/12
Very Large	(192, 96)	0.626	0.0293	0/12
Massive	(384, 192)	0.585	0.0458	1/12
Overall	pooled	0.445	0.000006	8/96

Table 1: Per-architecture and overall Pearson correlation between zero-shot diagonal strength and transfer advantage across 12 MNIST rotation angles ($n = 12$ per architecture, $n = 96$ pooled). All 8 architectures individually significant at $p < 0.05$. The overall pooled correlation is lower than individual correlations because pooling across architectures with different base accuracies adds variance to the relationship.

Table 1 shows the per-architecture breakdown. All eight architectures produced statistically significant individual correlations, ranging from $r = 0.585$ to $r = 0.822$.

The best transfer outcome in the rotation set was the Medium architecture at 15 degrees (diagonal strength = 0.839, transfer advantage = +0.006). The worst was the Tiny architecture at 180 degrees (diagonal strength = 0.222, transfer advantage = -0.073). High diagonal strength corresponds to small rotations that preserve digit structure, and low diagonal strength corresponds to extreme rotations where source features no longer apply.

A slight weakening of individual architecture correlations at larger sizes (Massive $r = 0.585$ vs Medium $r = 0.822$) is a real pattern in the data. One possible explanation is that larger architectures extract more rotation-invariant features, which would partially decouple zero-shot performance from fine-tuning outcome. This remains an open question.

5.3 Blur Results

Table 2 summarizes the blur validation results across both datasets. Architectures used were Small (32, 16), Medium (64, 32), Large (128, 64), Very Large (192, 96), and Huge (256, 128), as described in Section 4.

Experiment	Correlation	P-value	N	Positive Transfers
MNIST blur	+0.495	0.012	25	14/25
Fashion-MNIST blur	+0.564	0.003	25	3/25

Table 2: Pearson correlation between zero-shot diagonal strength and transfer advantage for motion blur experiments (3×3 to 11×11 kernel sizes). Five architecture sizes tested across five blur levels per dataset ($n = 25$ per experiment): Small (32, 16), Medium (64, 32), Large (128, 64), Very Large (192, 96), Huge (256, 128).

Both blur experiments produced positive correlations consistent with the rotation result, confirming that the binary prediction pattern holds across two geometric transform types. The Fashion-MNIST blur correlation ($r = 0.564$) was the strongest binary result in this paper, despite Fashion-

MNIST being harder than MNIST for the tested architectures. Blur succeeds as a geometric transform on Fashion-MNIST where rotations do not, because blur preserves clothing texture structure while rotations destroy it.

5.4 Boundary Conditions

Fashion-MNIST rotations were abandoned mid-test because all observed transfers were negative before the test completed. Texture-based features in clothing images do not survive rotation in the same way that edge-based features in digit images do. Additional boundary conditions involving non-geometric shift types are discussed in Section 9.

6 Magnitude Prediction on Degraded Data

6.1 Setup

We test whether zero-shot diagonal strength predicts the magnitude of transfer benefit on structure-destroying degradation shifts. In contrast to the geometric shift setting, the question here is not whether transfer will help but how much. Clean pre-training was positive in nearly all degradation cases, so the outcome variable is the size of the transfer advantage rather than its sign.

Four degradation conditions were validated, covering three degradation types. Gaussian noise was tested on both MNIST and Fashion-MNIST. Salt-and-pepper noise and contrast reduction were tested on MNIST only. For each condition, five architecture sizes were tested across five severity levels, producing 25 architecture-severity pairs per experiment.

6.2 Results

Table 3 summarizes all four magnitude prediction experiments.

Experiment	Degradation Type	Correlation	P-value	N	Positive Transfers
MNIST Gaussian	Additive noise	-0.941	<0.00001	25	25/25
Fashion-MNIST Gaussian	Additive noise	-0.786	0.000003	25	23/25
MNIST Salt-Pepper	Impulse noise	-0.730	0.000034	25	25/25
MNIST Contrast	Quality reduction	-0.573	0.003	25	23/25

Table 3: Pearson correlation between zero-shot diagonal strength and transfer advantage across four degradation experiments. All correlations are negative, confirming the magnitude prediction regime: lower diagonal strength predicts larger transfer benefit.

All four experiments produced statistically significant negative correlations. The strongest was MNIST Gaussian noise ($r = -0.941$), where all 25 transfers were positive and the diagonal strength range was 0.223 to 0.925. The weakest was MNIST contrast reduction ($r = -0.573$), which still reached high significance ($p = 0.003$) and had the largest diagonal range of any experiment in this paper (0.105 to 0.945).

The inverse magnitude pattern replicated across additive noise, impulse noise, and quality degradation, showing that the effect is not specific to one corruption type.

6.3 Interpretation

A model with low F on the degraded target domain produces poor zero-shot performance because its source-domain features do not generalize effectively to heavily corrupted inputs. Fine-tuning that model on the degraded target data provides a large benefit because the pre-trained weights carry clean structural information that the degraded data alone could not provide. A model with high F is being evaluated on lightly degraded data that already resembles its source domain, so fine-tuning provides less incremental gain.

This interpretation is consistent with the observation that all 25 transfers were positive in the MNIST Gaussian and salt-and-pepper experiments. When the target data is degraded, clean pre-training always provided something useful in the tested settings. The question is only how much.

6.4 Boundary Conditions

Fashion-MNIST salt-and-pepper noise produced $r = -0.392$ with $p = 0.053$, just below the significance threshold, and was excluded from the validated results. The diagonal range in that experiment was 0.270, below the 0.3 minimum observed across validated experiments. Fashion-MNIST appears to require more diagonal variation than MNIST to produce reliable magnitude predictions, possibly because its lower base accuracy compresses the operating range of the predictor.

7 Temporal Regime-Shift Validation on Financial Data

The image experiments establish the two-regime prediction pattern within computer vision. We report one validation on tabular financial data to test whether the same binary prediction rule produces a correct directional prediction outside of image classification.

7.1 Setup

We used the Lending Club loan dataset (Lending Club, 2016), treating the 2013–2014 period as the source domain and the 2015–2016 period as the target domain. Unlike the image experiments, which use scikit-learn MLPClassifier, the financial validation used a separate PyTorch feed-forward network with class weighting to address class imbalance (paid = 0.60, default = 2.96). The model was trained on 370,039 source-period loans (16.9% default rate).

The model was then evaluated zero-shot on the 2015–2016 period (852,607 loans, 17.7% default rate) and diagonal strength was computed from the resulting confusion matrix. Fine-tuning was run on 682,085 target-period training loans for 20 epochs. A scratch model was trained on the same target data for 40 epochs. Transfer advantage was computed on 170,522 held-out test loans.

For this temporal regime-shift check, we applied the same binary decision rule used in the image-domain binary experiments, treating diagonal strength below 0.7 as a negative-transfer indicator. This threshold was not fitted to the financial data. It was carried over as a heuristic decision rule from the image-domain binary experiments and should not be interpreted as a validated threshold for tabular financial data.

7.2 Result

Zero-shot diagonal strength on the 2015–2016 data was 0.6731, below the 0.7 threshold. The prediction was therefore negative transfer. The actual fine-tuned accuracy was 0.6233 and the scratch accuracy was 0.6450, producing a transfer advantage of -0.0217. The prediction was correct.

Statistical significance of the difference between fine-tuned and scratch performance was tested via z-test on proportions over the held-out target-period test set: $z = 13.24$, $p < 0.000001$, $n =$ 170,522 loans. A full data integrity audit of the financial experiment confirmed 31 of 31 checks passed.

7.3 Scope

This is a single confirmed directional prediction on one dataset with one temporal regime shift. It is not a correlation study and does not establish the same level of evidence as the image experiments. We report it as temporal regime-shift applicability evidence only. Whether the prediction rule generalizes to other financial datasets, other temporal shifts, or other tabular domains is not established by this result.

8 Off-Diagonal Entropy as a Secondary Signal

8.1 Definition

Off-diagonal entropy measures how randomly the model distributes its mistakes across incorrect classes in the zero-shot confusion matrix. It is computed from the normalized off-diagonal mass of the confusion matrix after removing the diagonal entries. Higher off-diagonal entropy means the model's mistakes are more evenly spread across all incorrect classes. Lower off-diagonal entropy means the model's mistakes concentrate on a small number of incorrect classes.

8.2 Result

In the MNIST rotation experiments, we computed the Pearson correlation between off-diagonal entropy and transfer advantage at two sample sizes.

At $n = 12$ (single architecture, 12 rotation angles), the correlation was $r = -0.507$ with $p = 0.092$. This did not reach the significance threshold.

At $n = 96$ (eight architectures, 12 rotation angles each), the correlation was $r = -0.313$ with $p = 0.002$. This reached significance.

The signal reached significance at the larger sample size under the same metric definition and analysis procedure. In this setting, the $n = 12$ comparison did not detect a signal that became significant at $n = 96$, suggesting that small samples can miss a real effect for this metric.

8.3 Interpretation and Scope

Off-diagonal entropy and diagonal strength measure structurally different things. Diagonal strength captures how often the model is correct. Off-diagonal entropy captures how structured or random the model's mistakes are. A model that concentrates its mistakes on a small number of confusable classes may carry more transferable structure than a model whose mistakes are spread evenly across all incorrect classes.

The entropy signal is substantially weaker than diagonal strength ($r = -0.313$ vs $r = 0.445$ at $n = 96$) and is best treated as an exploratory secondary finding until it receives broader multi-domain validation.

9 Discussion and Limitations

9.1 What This Work Shows

The central finding is that the type of domain shift determines how zero-shot diagonal strength should be interpreted as a transfer predictor. On geometric shifts, higher diagonal strength predicts a more favorable transfer outcome. On degradation shifts, lower diagonal strength predicts a larger transfer benefit. The same scalar, from the same single forward pass, carries opposite predictive information in the two settings.

The binary prediction result held across all eight tested MLP architecture sizes in the rotation experiments and replicated on blur transforms across two datasets. The magnitude prediction result replicated across four degradation experiments spanning three degradation types and two datasets. Both regimes have documented boundary conditions where the method did not produce useful predictions, and those conditions are reported throughout the paper.

9.2 Boundary Conditions Not Covered in Earlier Sections

Two additional boundary conditions were encountered during validation.

CIFAR-10 (Krizhevsky, 2009) Gaussian noise produced a statistically significant correlation ($r = -0.478$, $p = 0.016$) but a diagonal range of only 0.119 across the five tested architectures. This range is too narrow to support reliable prediction in the tested setting. Whether larger or more diverse architecture sets would produce a usable diagonal range on CIFAR-10 is not established.

The KMNIST (Clanuwat et al., 2018) to MNIST cross-domain pair failed entirely. The diagonal range was 0.026 across five tested architecture-condition pairs and no usable correlation was found. The near-zero feature overlap between Japanese and Arabic handwriting appears to have removed any useful predictive signal from zero-shot diagonal strength in that setting. This suggests that the method requires some structural similarity between source and target domains to produce a useful signal.

9.3 Limitations

Small sample sizes in blur, noise, and contrast experiments. The five-architecture experiments ($n = 25$ per condition) are sufficient for statistical significance but small in absolute terms. The rotation experiment ($n = 96$) is the most reliable evidence base in this paper.

MLP architectures only for the rotation universality claim. The result that all eight architecture sizes individually produced significant correlations was established on fully-connected networks only. Whether the same architecture-independence holds for convolutional networks on geometric shifts is not tested here.

Image and financial domains only. All primary experiments use image classification benchmarks. The financial validation is a single directional prediction. Whether the two-regime taxonomy applies to NLP, audio, or other tabular domains is unknown.

The 0.7 threshold was not cross-validated. The binary prediction threshold of 0.7 used in the financial validation was carried over from image experiment observations, not fitted to a held-out set. Its reliability as a decision boundary in other settings is not established.

Training budget matching. The fine-tuned model receives 20 source-domain epochs plus 20 target-domain epochs, while the scratch model receives 40 target-domain epochs. Whether source-domain exposure counts equivalently to target-domain exposure for learning purposes is not tested.

9.4 Future Work

The most direct extension is to test the two-regime taxonomy on convolutional networks across a wider range of geometric and degradation shift types. The rotation universality result was established on MLPs and whether it holds for CNNs is an open question.

A second direction is to test whether the magnitude prediction pattern holds on Fashion-MNIST salt-and-pepper noise with a larger architecture set, which would clarify whether the $p = 0.053$ near-miss reflects a genuine boundary condition or an insufficient sample.

A third direction is to validate the binary prediction rule on additional financial and tabular datasets with temporal regime shifts, to determine whether the single financial result generalizes or was specific to the Lending Club temporal shift.

Finally, off-diagonal entropy warrants multi-domain validation to determine whether it provides independent predictive signal beyond what diagonal strength already captures, and whether the sample-size dependency observed here holds across other experimental settings.

Code Availability

The diagonal strength predictor implementation, all validation scripts, stored JSON result files, and the complete audit certification are available at the GitHub repository Wise314/identity-framework-extensions.

References

LeCun, Y., Bottou, L., Bengio, Y., and Haffner, P. Gradient-based learning applied to document recognition. *Proceedings of the IEEE*, 86(11):2278–2324, 1998.

Xiao, H., Rasul, K., and Vollgraf, R. Fashion-MNIST: A novel image dataset for benchmarking machine learning algorithms. *arXiv preprint arXiv:1708.07747*, 2017.

Lending Club. Loan data 2013–2016. Available at `https://www.kaggle.com/datasets/wordsforthewise/lending-club`, 2016.

Krizhevsky, A. Learning multiple layers of features from tiny images. Technical Report, University of Toronto, 2009.

Clanuwat, T., Bober-Irizar, M., Kitamoto, A., Lamb, A., Yamamoto, K., and Ha, D. Deep learning for classical Japanese literature. *arXiv preprint arXiv:1812.01718*, 2018.

Pan, S. J. and Yang, Q. A survey on transfer learning. *IEEE Transactions on Knowledge and Data Engineering*, 22(10):1345–1359, 2010.

Yosinski, J., Clune, J., Bengio, Y., and Lipson, H. How transferable are features in deep neural networks? In *Advances in Neural Information Processing Systems (NeurIPS)*, 2014.

Nguyen, C., Hassner, T., Seeger, M., and Archambeau, C. LEEP: A new measure to evaluate transferability of learned representations. In *Proceedings of the International Conference on Machine Learning (ICML)*, 2020.

Tran, A. T., Nguyen, C. V., and Hassner, T. Transferability and hardness of supervised classification tasks. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 2019.

Ben-David, S., Blitzer, J., Crammer, K., Kulesza, A., Pereira, F., and Vaughan, J. W. A theory of learning from different distributions. *Machine Learning*, 79(1–2):151–175, 2010.

Patent Disclosures

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